**AGENDA**

Student: Theresia Karyna- theresiakaryna@gmail.com

Mentor: Carla Gomes - carladcpn@gmail.com
Meetings: every other Monday at 9:30am (Carla) / 7:30pm (Karyna)

## March 31st

**Meeting Summary**

Karyna is developing a Capstone project focused on ADHD, aiming to build a system that uses brain data (EEG), sleep patterns, and physiological signals (e.g., heart rate) to understand stress and cognitive states. The goal is to process this data using AI and generate personalized suggestions to improve quality of life, with a feedback loop that allows the system to learn and improve over time.

She has a strong technical foundation, including experience with electronics (soldering, PCBs) and programming (Python and R). While she is interested in building a device, she is open to starting with existing datasets and focusing on software development—specifically training an AI model to receive, process, and interpret data before moving into hardware.

Karyna referenced a therapy-related concept ("Crispin," possibly CRISPR or another method) and is currently working on an article under peer review, though its acceptance is still uncertain. She has already begun organizing her work in a Google document.

At this stage, she plans to:

Work with available data (EEG, sleep, stress)
Build a system that translates data into actionable suggestions
Train the AI model to improve suggestions based on user feedback
Explore which AI/LLM tools would be most appropriate

In terms of mentoring, Karyna prefers structured guidance with clear next steps. She benefits most from regular check-ins, process-oriented discussions, and direct, detailed feedback on what is working well and what can be improved.

**Next Steps**
Develop Phase 1: Problem Understanding and Research
Prepare a clear summary of findings (research + gaps in existing solutions)
Present Phase 1 progress in the next meeting

## April 13th

**Karyna explained the findings of her research.**

**We developed the Project Concept (One pager).**

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| **Finally we talked about Phase 3.**<br><br>https://canva.link/gl345q3bzwjum6s |
| **Next Steps**<br>Phase 3 |

## May 18th

Agenda:

**1- Ice breaker:** What's something you enjoy doing that most people wouldn't guess about you?

**2- Revise project description and Timeline**
Highlight in green the 3 most important ideas of the project.

a) What problem is the app solving?
b) How does the app work?
c) What makes this project useful or different?




**3- Karyna presents her research on Phase 3**.

**4- Carla comments the research** (strong points and weak points)

**5- Discussion:** How to create a Table with Core/ Context Variables?

Variable Categories

- Load / Trigger Variables: Variables that may increase cognitive overload or make ADHD symptoms worse. These represent demands, distractions, or stressful situations that affect the user.
- State Variables: Variables that reflect the user's current mental or cognitive state. These help the app understand how the user is functioning at the moment.
- Recovery / Regulation Variables: Variables related to recovery, regulation, or coping strategies. These may help reduce overload and improve focus or emotional regulation.
- Personalization Variables: Variables that help identify individual patterns over time. These allow the app to become more personalized and adaptive for each user.

| Type | Variable | Why is it important? |
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| Load / Trigger | Task difficulty | Higher cognitive demand may increase overload |
| State | Stress level (overwhelm) | Indicates possible cognitive overload. This data may help the app associate an activity or workload amount with an overwhelming feeling. |
| Regulation Variable | Sleep quality | Good sleep improves recovery and focus |
| State | Fatigue | Fatigue may reduce attention and motivation |
| Load/trigger, recovery/regulation | Current activity | Activity may provide context on why the user is experiencing a state.<br>If the activity is associated with stress, it may become a trigger (e.g., homework).<br>If the activity is associated with happiness, it may become a regulatory strategy (e.g., reading). |
| Load/trigger | Workload | The amount of work (or impending work) may provide context on the worried state of the user. Many users may feel like there is too much going on, so they are stuck and feel unable to do anything. |
| State | Focus/attention | Increased focus may allow doing heavier tasks, and decreased focus may call for a regulation strategy. |
| Load/trigger, regulation | Environment **type/state** | A stress-inducing environment, such as a hospital or a busy place, may cause increased stress (even though the user may not be realising it). This may help the app understand external variables to the person and associate it with the user's state (and give better suggestions).<br>A good environment or a preferred environment by the user, such as a library, may help the app associate certain places with a calmed state and increased focus, helping it give better suggestions. |
| Recovery/regulation | Medications | Medication intake may help with focus and sleep quality, becoming a solution for some states. |
| Load/trigger, Personalization | Productive times | This may help the app give better suggestions, moving the heavier tasks to more productive times. |
| State | Sleepiness | The app may suggest rest instead of more work during times of sleepiness. |
| State | Motivation | Is the user motivated to a certain activity? This may help an app associate an activity with increased motivation, giving suggestions to increase motivation (and help the user finish tasks). |

**Next Steps**

## Phase 2: Existing Solutions

Summary:

1. **No System Combines All Important Data:** Most ADHD apps only track one thing, such as habits, schedules, medication, sleep, or brain activity. There is currently no common system that combines brain signals (EEG), heart rate, stress levels, and sleep data into one complete picture of a person's ADHD symptoms. **Opportunity:** Create a system that combines all these data sources to better understand a person's focus, stress, and overall ADHD condition.

2. No Truly Personalized AI Advice: Many apps provide the same reminders, lessons, or activities to every user. Even when they collect data, they usually only show graphs or statistics instead of giving personalized recommendations. **Opportunity:** Use AI to analyze the user's data and provide customized suggestions, such as when to take a break, how to reduce stress, or the best time to focus on important tasks.
3. Limited Ability to Learn From the User : Most ADHD tools follow fixed programs and do not change much based on whether a user improves or struggles. **Opportunity:** Develop a system that learns from the user's behavior and adjusts its recommendations over time based on what works best for that individual.
4. **No Real-Time Stress or Overload Detection:** Some wearable devices can measure stress-related signals, but they usually only monitor them rather than predicting when a person is becoming overwhelmed or mentally exhausted. **Opportunity:** Build a system that detects signs of stress, overload, or attention loss in real time and provides support before problems become severe.
5. **High Cost and Limited Accessibility:** Many advanced ADHD technologies are expensive, require prescriptions, or depend on specialized equipment. Some are designed only for children or only for adults. **Opportunity:** Create a more affordable system that works with common consumer devices and can be used by people of different ages.
6. **Limited Scientific Evidence:** Many existing ADHD apps and brain-training tools have not been tested through large, long-term scientific studies. Some claims about their effectiveness remain uncertain. **Opportunity:** Develop a system that is based on scientific research, uses transparent AI methods, and measures whether users actually improve over time.
7. **No Long-Term Quality-of-Life Focus:** Most tools focus on a single area, such as attention, organization, sleep, or medication. They do not track how these factors affect a person's overall daily life and well-being over long periods. **Opportunity:** Create a platform that connects sleep, stress, brain activity, productivity, and daily functioning to provide long-term support and improve quality of life.

| Existing Solution | Is it an App? | Goal | Input | Output | Differentiation | Gap |
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| Inflow | Yes | Help adults manage ADHD symptoms through CBT-based self-management<br><br>CBT = cognitive behavioural therapy e.g. task chunking (breaking tasks down into smaller steps) | Self-reported symptoms, user reflections, task/habit input | CBT exercises, psychoeducation, habit-building modules, progress support, accountability coaching | Combines CBT, psychoeducation, social support, and in-app accountability coaching in one ADHD-focused app (1.1, 1.7) | No physiological or EEG input; no real-time stress/cognitive-state detection; limited AI personalization; evidence is mainly feasibility/usability, not strong replicated RCTs (1.1, 1.6) |
| Tiimo | Yes | Support planning, time management, and routines for neurodivergent users | Manual schedule entry, tasks, routines, reminders | Visual timelines, reminders, routine support | ADHD-friendly visual planner with simple, customizable routine design | No biometric/physiological input; no overload or stress detection; no AI-driven personalization; no adaptive feedback loop |
| Forest | Yes | Improve focus by reducing phone distraction | User-set focus timer | Virtual tree growth, gamified reward for staying off phone | Highly intuitive gamified phone-avoidance mechanic that makes | Not ADHD-specific; no physiological monitoring; no cognitive-state awareness; no |

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| | | | | | focus visible and rewarding (4.1) | personalized suggestions; no clinical ADHD evidence (4.1, 1.4) |
| Focus @Will | Yes | Improve concentration and productivity through tailored music | User preferences, proprietary AI assessment, timer settings | Curated focus music, pomodoro/focus timers, productivity tracking | AI-curated music plus timer-based productivity workflow for focus sessions (1.5, 1.4) | No peer-reviewed ADHD-specific evidence; no biometric input; no real-time physiological monitoring; limited adaptation beyond music selection (1.4) |
| EndeavorRx | Yes | Treat pediatric ADHD-related inattention through a prescription video game | Game-based attention/multitasking performance | Attention training tasks, gameplay progress, parent-facing companion tracking | FDA-authorized digital therapeutic with strongest RCT support among reviewed tools; directly targets sustained/selective attention (1.3, 1.2) | Children only; prescription access; primarily targets inattention, not hyperactivity/impulsivity; no physiological input; no stress detection; no daily-life personalized suggestions; likely expensive/inaccessible for some users (1.3, 1.4) |
| Apple Watch | Yes | General wellness tracking: activity, sleep, heart metrics | Accelerometer, gyroscope, optical heart-rate sensor, smartwatch behavioral data | HR, HRV, sleep/activity summaries, wellness metrics | Widely adopted wearable with strong consumer ecosystem and useful passive physiological data capture (8.1, 2.2) | Not ADHD-specific; no EEG; no direct cognitive-state inference; no ADHD-tailored interpretation or suggestions; mostly provides raw/consumer wellness metrics rather than actionable ADHD guidance (2.3, 8.1) |
| Oura Ring | Yes | Track sleep, recovery, and nocturnal physiology | Multi-wavelength PPG, accelerometer, temperature sensor | Sleep/wake estimates, sleep stages, HR, HRV, readiness/recovery metrics, temperature trends | High nocturnal HR accuracy vs ECG and low-burden ring form factor; strong sleep/wake detection (5.1, 6.1) | Sleep staging remains limited (~53% in some clinical contexts); no EEG; no ADHD-specific interpretation; no cognitive-state monitoring; no ADHD-tailored actionable suggestions (7.1, 5.2) |
| Muse Headba | Yes | Meditation support and consumer | 4–7 channel EEG signals | Real-time meditation/neurofee | Consumer-accessible EEG with real-time | Limited channels relative to clinical EEG; not well |

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| nd | | neurofeedback | | dback audio, calm/active brain-state indicators, session feedback | neurofeedback and home usability (3.1, 3.2) | validated for ADHD treatment; mainly meditation-focused; little multimodal integration with HR/sleep; disputed effectiveness in meta-analysis; no personalized ADHD management suggestions (3.1, 3.2) |
| Mightier | Yes | Teach emotional self-regulation using biofeedback games | Heart-rate sensor/wristband | Games adapt to arousal/heart rate, calming skill practice, emotion-regulation feedback | Real-time HR biofeedback embedded in engaging games for self-regulation learning (9.4) | HR only; no EEG; focused more on emotional regulation than ADHD cognitive focus; pediatric orientation; no sleep/stress-pattern analysis across time; no personalized daily-life recommendation engine (9.4) |
| FOCUS ADHD App | Yes | Improve medication adherence and monitor treatment | Self-reported medication intake, symptom ratings, treatment-plan data | Pill reminders, adherence logs, symptom trends, adverse-event logs, incentive-linked engagement | Medication-management focus with adherence tracking and optional incentives (1.1, 1.2) | Narrow use case; no physiological data; no cognitive-state monitoring; no behavioral personalization; adherence benefit depended on monetary incentives (1.1) |
| NeuroPlus | Yes | Improve attention through EEG neurofeedback and game-based training | EEG headset data during training | Neurofeedback games, attention-training sessions, real-time feedback | Combines neurofeedback with gamified exercises to make EEG training more engaging (9.1, 3.1) | No multimodal integration with sleep/HR/stress; limited validation; no broader daily-life recommendation system; no longitudinal personalized lifestyle guidance (9.1, 3.3) |
| DETEC-ADHD | Yes | Detect/classify ADHD and subtype using ML + EEG | Personal/medical data, EEG theta/beta ratio and related features | ADHD classification, subtype identification, web-based diagnostic output | ML-driven diagnostic web app combining EEG biomarkers with personal data (11.2, 11.1) | Diagnostic only; not an ongoing management tool; no intervention guidance; no adaptive feedback loop; no continuous monitoring; no |

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| | | | | | | sleep/stress integration (11.2, 11.3) |
| CogniFit | Yes | Deliver adaptive cognitive training exercises | Cognitive assessment and game/task performance | Personalized training activities, difficulty adjustment, progress tracking | ML-personalized cognitive training that adapts task difficulty over time (9.3, 9.4) | No physiological data input; no real-time biometric monitoring; no ADHD-specific overload detection; no wearable integration; unclear ADHD-specific validation strength (9.3, 9.2) |

## June 1st

**1- Ice breaker:** If you could spend one month shadowing any engineer, scientist, or inventor in the world, who would you choose and why?
Feng Zhang: Neuroscientist: Crisp autism -

**2- Karyna presents her findings on Phase 2- existing solutions**

**3- Discussion/Clarifying questions:**

- What do you now believe is the real gap your project is trying to solve?
- Which feature or idea do you think makes your app meaningfully different from a generic productivity or ADHD app?
- The application will grow together with the user. The suggestion will be personalized and grow along with the person. A lot of people receive feedback that is generic, but Karyna's will really understand the user's needs.

**4- Initial Ideation about Phase 4**: Define App Logic & Recommendations (INTELLIGENCE OF THE APP)
The idea is to finalize Parts A, B and C of Phase 4:

**Next Steps**
**Next Meeting: June 18th - Thursday**

### Phase 4 (parts A, B and C)

1. Step A: variables list divided into input and context variables
2. Step B: may use AI to help
    a. find literature and sources to identify states correctly
    b. list all possible states
    c. make a table with columns: Inferred State, Input Variables Involved, Context Variables Involved, Research Basis, Possible Interpretation, Recommendation
3. Step C: assign a recommendation for each state
4. Optional: feedback mechanism
    a. detect next state

b. new input within loop: did the previous recommendation work? yes/no → new variable. If statements, then change recommendations
   i. give same recommendation with another layer
   ii. different recommendation
   iii. same recommendation

## June 18th

**1- Ice breaker:** If you could instantly become an expert in one skill or subject, what would it be and why?
Skill: charismatic - comedian - make jokes

**2- Karyna presents her findings on Phase 4- (parts A, B and C):** Define App Logic & Recommendations (INTELLIGENCE OF THE APP)

***SYSTEM ARCHITECTURE:***

***OPTION 1:*** Consider all the combinations

INPUT + CONTEXT → COGNITIVE STATE → RECOMMENDATION

Entry A: INPUTS (around 5- considering grades 1-5)= $5^5$ = 3.125 combinations

Entry B: CONTEXT (around 5- considering 5 options for each)= $5^5$ = 3.125 combinations

This would give 3125 × 3125= 9.765.625 possibilities (NOT VIABLE)

This approach would require creating an enormous number of decision rules and is therefore not realistic.

***OPTION 2:*** Use Patter Recognition

INPUTS + CONTEXT → PATTERN RECOGNITION → COGNITIVE STATE → RECOMMENDATION

Instead of analyzing millions of possible combinations individually, the system groups similar patterns into a small number of meaningful cognitive states. (VIABLE)

PROBLEM: What if there is a combination of Inputs not considered?
The system is not expected to cover every possible scenario. Instead, it should recognize the most common and scientifically supported patterns. When an unfamiliar combination occurs, the system can default to a generic support pathway and refine its classification as more user data becomes available.

**3- Carla revises the system architecture:**

A) Define the Input Variables (internal conditions reported by the user): stress, focus, fatigue, etc.

B) Define the Context Variables (external or situational factors that may influence the user's cognitive state). Ex: current activity, task difficulty, etc.
C) (Optional for future versions) Define Derived Variables — variables learned by the system through historical data, such as recurring overload periods, productive times, or preferred regulation strategies.
D) Define the main Cognitive States the app should identify and provide a clear definition for each one. Ex: cognitive overload, motivation deficit, etc.
E) For each Cognitive State, identify the most likely Input Variables and Context Variables associated with it.
F) Create a Cognitive State Detection Table showing (maybe 2 - 3 rows with different combinations for the same Cognitive State)

| Cognitive State | Definition | Input Variables | Context Variables | Research Basis |
|---|---|---|---|---|

G) Define the Recommendation Library by identifying possible interventions for each Cognitive State.

The goal is not to cover every possible combination, but to identify the most meaningful cognitive patterns and connect them to evidence-based recommendations.

**4- Next Phase**

- PHASE 5: Design Screens & User Experience
- Please read this document for details:
- Let's use Figma, which is a design and prototyping tool used to create the visual interface and user experience of an app before development. It allows designers to build screens, map user flows, and create interactive prototypes without writing code.

**Next Steps**

PHASE 5

At the end of this phase, you will design:

**Screen 1** – Home Screen (Start Check-In and View Insights)

**Screen 2** – Check-In Screen (Stress, Focus, Fatigue, Motivation, and Context Questions)

**Screen 3** – Cognitive State Results Screen (Display the detected Cognitive State)

**Screen 4** – Recommendations Screen (Personalized recommendations based on the detected state)

**Screen 5** – Feedback Screen (Allow users to indicate whether the recommendation was helpful)

**Screen 6** – Insights Dashboard (Focus trends, stress trends, productive times, and recurring patterns)

You will then connect these screens into a clickable prototype that simulates the user experience from opening the app to receiving recommendations and reviewing personal insights.